ic_autolab round 12 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T04:45:38Z at git `fdeed51f64bdb7c43a639fdbef55d3ffeebbcdd6`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-4afdff` (`python3 tools/pcc/pcc.py standing PCC-20261104-4afdff`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**no_challenger**	none (best candidate `ksum4s-n12`, train S 7.413 / 8.176 / 15.041)	-	-	-	-	-	6.857 / 7.359 / 11.877	1.238 / 1.350 / 2.358	inc: 5.5x / 5.5x / 5.0x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

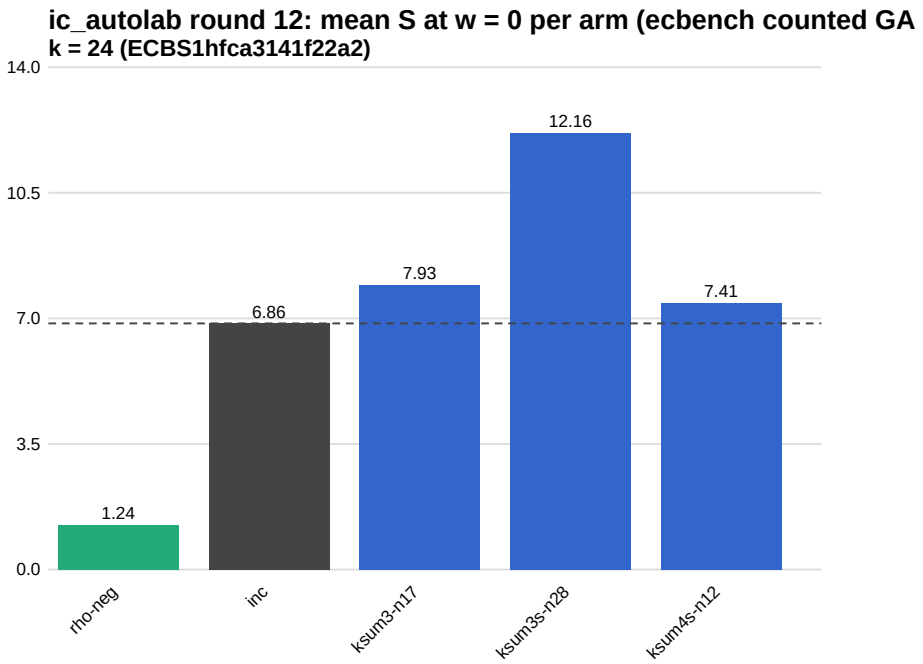
- k = 24: no_challenger

Train (development screen, in sample)

k = 24: session `ECBS1hfca3141f22a2` (run exit 0, verify exit 0, receipt sha256 `4c4b2bb576a6cccb0160ec4d99852cb81280d99ecd65c00169bf80786e8c6960`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.238 / 1.350 / 2.358	matched reference
`inc`	ksum4	12 `prime-projected-abscissa:size=12,scan_cap=192`	`ksum:m=4`	32	32	6.857 / 7.359 / 11.877	incumbent
`ksum3-n17`	ksum3	17 `prime-projected-abscissa:size=17,scan_cap=272`	`ksum:m=3`	32	32	7.931 / 8.861 / 17.222	alternative ksum3 (best S(0) 7.824 at N = 20): N = 0.85 x 20
`ksum3s-n28`	ksum3s	28 `prime-projected-abscissa:size=28,scan_cap=448`	`ksum:m=3`	32	32	12.164 / 12.642 /	exploration: family

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
						16.940	ksum3s untried at k = 24, N = 0.344 x N_a
`ksum4s-n12`	ksum4s	12 `prime-projected-abscissa:size=12,scan_cap=192`	`ksum:m=4`	32	32	7.413 / 8.176 / 15.041	exploration: family ksum4s untried at k = 24, N = 0.148 x N_a



train screen

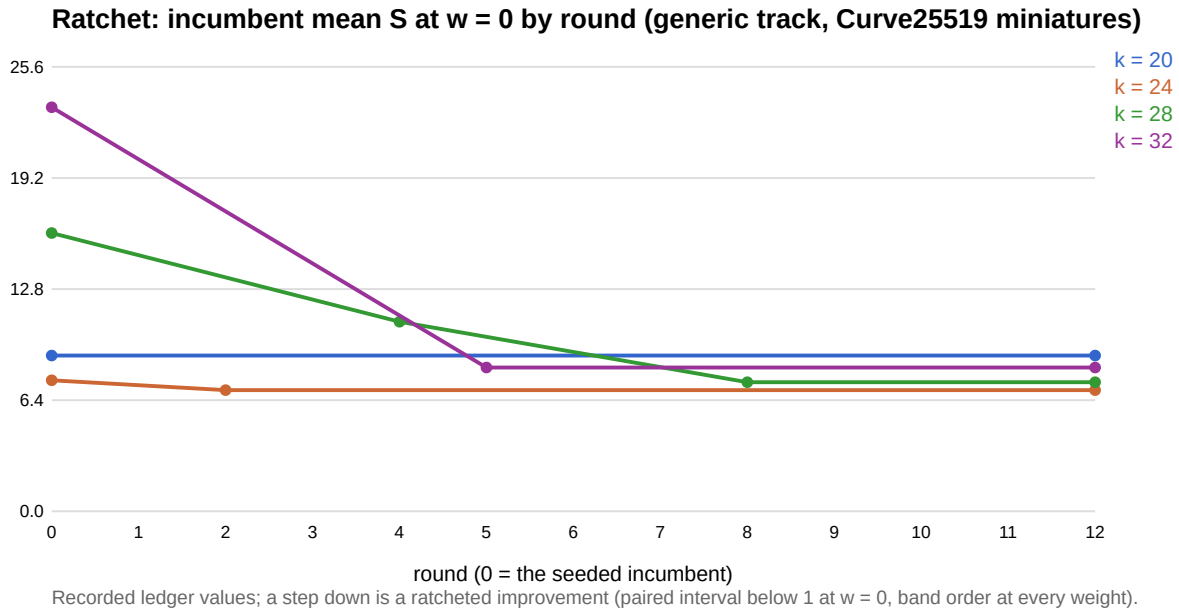
Selection

- k = 24: no_challenger (Pareto alternative at w = 1: `ksum4s-n12`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`ksum3-n11` (`prime-projected-abscissa:size=11,scan_cap=176` + `ksum:m=3`)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	`ksum4-n12` (`prime-projected-abscissa:size=12,scan_cap=192` + `ksum:m=4`)	2	6.985 / 7.514 / 12.271	6.0x / 5.9x / 5.6x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
28	`ksum4-n18` (`prime-projected-abscissa:size=18,scan_cap=288` + `ksum:m=4`)	8	7.445 / 7.910 / 12.100	7.8x / 7.6x / 6.5x
32	`ksum4-n26` (`prime-projected-abscissa:size=26,scan_cap=416` + `ksum:m=4`)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.